

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1iISFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

CSS

mactime -b body.txt > timeline.txt

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

```
(abhi@kali)-[~]
└─$ ls evidence.img
v/v 3270339: $MBR
v/v 3270340: $FAT1
v/v 3270341: $FAT2
V/V 3270342: $OrphanFiles
└─$ r/r 5: test.txt
zsh: no such file or directory: r/r
└─$ lstat evidence.img 5
Invalid metadata address (fatxxfs_inode_lookup: 5 is not an inode)
└─$ icat evidence.img 5 > recovered.txt
Invalid metadata address (fatxxfs_inode_lookup: 5 is not an inode)
└─$ cat recovered.txt
└─$ ls -m / -r evidence.img > body.txt
└─$ mactime -b body.txt > timeline.txt
Old package separator ... deprecated at /usr/bin/mactime line 154.
Old package separator ... deprecated at /usr/bin/mactime line 167.
└─$ cat timeline.txt
└─$
```

```
zsh: corrupt history file /home/abhi/.zsh_history
└─$ dd if=/dev/zero of=evidence.img bs=1M count=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 1.97323 s, 53.1 MB/s
└─$ mkfs.vfat evidence.img
mkfs.fat 4.2 (2021-01-31)
└─$ fdisk evidence.img
FILE SYSTEM INFORMATION
File System Type: FAT16
OEM Name: mkfs.fat
Volume ID: 0x843a3afe
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory):
File System Type Label: FAT16
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 204799
* Reserved: 0 - 3
** Boot Sector: 0
* FAT 0: 4 - 203
* FAT 1: 204 - 403
* Data Area: 404 - 204799
** Root Directory: 404 - 435
** Cluster Area: 436 - 204799
METADATA INFORMATION
Range: 2 - 3270342
Root Directory: 2
CONTENT INFORMATION
Sector Size: 512
Cluster Size: 2048
Total Cluster Range: 2 - 51092
FAT CONTENTS (in sectors)
```

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. **Archive Evidence:**

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. **Store Securely:**

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result: Thus, the disk image was successfully analyzed using The Sleuth Kit to identify file system information and recover relevant files for forensic analysis.